

Object Oriented Programming Lab

Lab 12

Instructions

Work on this lab individually. You can use your books, notes, handouts etc but you are not allowed to borrow anything from your peer student.

Write down the lab number, task number, your name and roll number at the start of each task code using comments.

For Example

```

/*****
Lab:    xx
Task:   yy
Developer:
        Roll #: XXXYYYZ000
        Name:  XYZ
*****/

```

What you have to do

Program the following tasks in C++, compile and execute them.

Task

You are required to design a shape hierarchy system using **inheritance** and **polymorphism**. Create base class **Shape** with following functions:

- A virtual function `draw()`
- A non-virtual function `info()` to show the difference in binding

Create Three Derived Classes:

1. Circle
2. Rectangle
3. Triangle

Each class must inherit publicly from **Shape**

- Override the `draw()` function (polymorphic).
- Optionally override the `info()` function (but calls will still be statically bound).

In `main()`

1. Create an **array of Shape*** of size 3.
2. Store in it dynamically created objects:
 - `shapes[0] = new Circle();`
 - `shapes[1] = new Rectangle();`
 - `shapes[2] = new Triangle();`
3. Loop through the array and call:
 - `shapes[i]->draw();` // must show dynamic binding
 - `shapes[i]->info();` // must show static binding
4. Delete the objects to avoid memory leaks.

Expected Output (Example)

```

Drawing Circle...
Shape Info (Base class) - Static Binding

```

```

Drawing Rectangle...
Shape Info (Base class) - Static Binding

```

```

Drawing Triangle...
Shape Info (Base class) - Static Binding

```

😊😊😊 **BEST OF LUCK** 😊😊😊